

Lecture 17 Feb 25th

Recall last time:

$$|\Psi(t)\rangle = \sum_m a_m(t) e^{-iE_m t} |m\rangle$$

Accounts for transitions
at some time in the
future.

$a_m(t) = \langle m | \Psi(t) \rangle$ always.

By using $i\hbar \frac{d}{dt} |\Psi(t)\rangle = \hat{H} |\Psi(t)\rangle$

$$\Rightarrow \boxed{\frac{da_m(t)}{dt} = -i \sum_k a_k(t) e^{-i\omega_{km}t} V_{mk}}$$

Where $V_{mk} = \langle m | \hat{V}(t) | k \rangle$

$$\omega_{km} = E_k - E_m$$

As before let us plug in a perturbative exp.

$$a_m(t) = a_m^{(0)}(t) + \lambda a_m^{(1)}(t) + \lambda^2 a_m^{(2)}(t) + \dots$$

Assume $|\psi(t=0)\rangle = |i\rangle$

$$e^{i\theta} = \cos\theta + i\sin\theta$$

$$\Rightarrow a_m = \delta_{mi}$$

Order by order in λ (Phrase as $i \rightarrow f$)

$$O(\lambda^0): \quad \frac{da_f^{(0)}(t)}{dt} = 0 \rightarrow \text{Nothing of } O(\lambda^0) \text{ on RHS.}$$

$$O(\lambda^1): \quad \frac{da_f^{(1)}}{dt} = -i \sum_k a_k^{(0)}(t) e^{-i\omega_{kf}t} V_{fk}$$

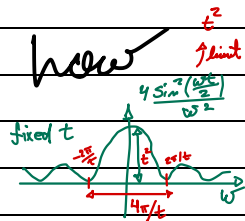
$$= -i \left(a_i^{(0)}(t) e^{-i\omega_{if}t} V_{fi} \right)$$

$$\Rightarrow a_f^{(1)} = -i \int_0^t dt' e^{-i\omega_{if}t'} V_{fi} \quad \text{--- } -i \sin \frac{\omega_{if}t}{2} \quad \parallel \frac{\omega_{if}}{2}$$

Assume $i \neq f$
and V constant

$$= V_{fi} e^{-\frac{i\omega_{if}t}{2}} \left(\frac{e^{-\frac{i\omega_{if}t}{2}} - e^{\frac{+i\omega_{if}t}{2}}}{\omega_{if}} \right)$$

If initially system was in state $|i\rangle$, how likely is it to find it in $|f\rangle$?



$$P_{i \rightarrow f}(t) \cong |a_f^{(1)}(t)|^2 = \lambda^2 |V_{fi}|^2 \frac{\sin^2\left(\frac{\omega_{if}t}{2}\right)}{(\omega_{if}/2)^2}$$

$$\underline{t\omega_{if} \gg 1} \rightarrow \lambda^2 |V_{fi}|^2 2\pi t \delta(\omega_{if}) \rightarrow \text{E conserv.}$$

$\lim_{t \rightarrow \infty} \frac{\sin^2(\omega t)}{t} \rightarrow \pi \delta(\omega)$

Fermi's Golden Rule (primitive version):

$$\Gamma_{i \rightarrow f} = \frac{P_{i \rightarrow f}(t)}{t} = 2\pi \lambda^2 |V_{fi}|^2 \delta(\omega_{if})$$

↳ t independent

Rate $\xrightarrow{\quad}$ Amplitude²

How about $|i\rangle$?

$$P_{ii} = |a_i(t)|^2 \approx |a_i^{(0)} + \lambda a_i^{(1)}(t)|^2$$

$$= |1 - i\lambda V_{ii}t| \rightarrow \text{phase shift piece.}$$

can be normalised away.

$\alpha(\lambda^2)$: Feb 27 Dec 18th

$$\frac{d a_f^{(2)}(t)}{dt} = -i \sum_k a_k^{(1)} e^{-i\omega_k t} V_{fk}$$

$$= -i \sum_k e^{-i\omega_k t} e^{-i\omega_k t} \frac{\sin(\frac{\omega_k t}{2})}{\omega_k/2} V_{fk} V_{ki}$$

$$= -i e^{-i\omega_i t} \sum_k \frac{\sin(\frac{\omega_k t}{2})}{\omega_k/2} V_{fk} V_{ki}$$

$$\Rightarrow a_f^{(2)}(t) = - \sum_k V_{fk} V_{ki} \int_0^t e^{-i\omega_k t'} \frac{\sin(\frac{\omega_k t'}{2})}{\omega_k/2} dt'$$

$a_i^{(2)}(t=0) = 0$

complicated, but $E_i \approx E_f$

Overall

Average

$$\Gamma_{i \rightarrow f} \cong 2\pi \left| \lambda V_{fi} + \lambda \sum_k \frac{V_{fk} V_{ki}}{\omega_{ik}} \right|^2 @ E_f \approx E_i$$

Usually many states with $E_f \approx E_i$,

so

$$\sum_{\text{continuum } f} |C_f|^2 \rightarrow \int dE_f \rho(E_f) |C_f|^2$$

$$\Gamma_i = \int d\Gamma_{i \rightarrow f} = \int 2\pi |\dots|^2 \rho(E_f) dE_f$$

of states between E_f and $E_f + dE_f$

Interaction picture $\hat{H}(t) = \hat{H}_0 + \lambda \hat{V}(t)$

$$|\psi(t)\rangle_I = e^{i\hat{H}_0 t} |\psi(t)\rangle_S = e^{-i\lambda \hat{V} t} |\psi(0)\rangle \xrightarrow[\substack{\lambda \rightarrow 0 \\ H(t) = H_0}]{\lambda \rightarrow 0} |\psi(0)\rangle$$

$$\begin{cases} \hat{A}_I = e^{i\hat{H}_0 t} \hat{A}_S e^{-i\hat{H}_0 t} \\ \hat{V}_I = e^{i\hat{H}_0 t} \hat{V} e^{-i\hat{H}_0 t} \end{cases}$$

$$\cancel{S} \quad \hat{H} = \hat{H}_0 + \lambda \hat{V} \Rightarrow |\psi(t)\rangle_I = e^{-i\lambda \hat{V} t} |\psi(0)\rangle$$

	Schrod.	Heisenberg	Interaction
Ket	$ \psi_S(t)\rangle = e^{i\hat{H}_0 t} \psi_S(0)\rangle$	Const	$ \psi(t)\rangle_I = e^{i\hat{H}_0 t} \psi(t)\rangle_S$
EV.	const	$A_H(t) = e^{i\hat{H}_0 t} A_S e^{-i\hat{H}_0 t}$	$A_I(t) = e^{i\hat{H}_0 t} A_S e^{-i\hat{H}_0 t}$

$$i\hbar \frac{\partial}{\partial t} |\psi(t)\rangle_I = -H_0 e^{iH_0 t} |\psi(t)\rangle_S + e^{iH_0 t} (H_0 + \lambda V) |\psi(t)\rangle_S$$

$$= \left(e^{iH_0 t} \lambda V e^{-iH_0 t} \right) e^{iH_0 t} |\psi(t)\rangle_S$$

$$\Rightarrow \boxed{i\hbar \frac{\partial}{\partial t} |\psi(t)\rangle_I = V_I |\psi(t)\rangle_I} \quad \left. \begin{array}{l} \text{Almost} \\ \text{Schrödinger} \\ \text{except} \\ H \rightarrow V_I \end{array} \right\}$$

BTW

$$\boxed{\frac{d}{dt} A_I = \frac{1}{i\hbar} [A_I, H_0]}$$

Almost Heisenberg, except $H \rightarrow H_0$

$$|\psi(t)\rangle_I = \sum_n C_n(t) |n\rangle \Leftrightarrow |\psi(t)\rangle_S = \sum_n C_n(t) e^{-iH_0 t} |n\rangle$$

$$\begin{aligned} i\hbar \frac{\partial}{\partial t} \langle n | \psi(t) \rangle_I &= \langle n | V_I | \psi(t) \rangle_I \\ &= \sum_k \langle n | V_I | k \rangle \langle k | \psi(t) \rangle_I \\ &= \sum_k V_{nk} e^{-i\omega_{nk} t} C_k(t) \end{aligned}$$

Same eq.

$$\Rightarrow \boxed{i\hbar \frac{d}{dt} C_n(t) = \sum_k V_{nk} e^{-i\omega_{nk} t} C_k(t)}$$

Linear system of equations:

$$i \begin{pmatrix} \dot{c}_1 \\ \vdots \\ \vdots \end{pmatrix} = \begin{pmatrix} V_{11} & V_{12} e^{-i\omega_{21}t} & \dots \\ V_{21} e^{-i\omega_{12}t} & \ddots & \\ \vdots & & \ddots \end{pmatrix} \begin{pmatrix} c_1 \\ \vdots \\ \vdots \end{pmatrix}$$

Closer look at the time evolution operator. We ought to be able to define

$$|\psi(t)\rangle_I = \hat{U}_I(t) |\psi(0)\rangle_I$$

Recall that

$$i \frac{\partial}{\partial t} |\psi(t)\rangle_I = \hat{V}_I(t) |\psi(t)\rangle$$

$$i \frac{\partial}{\partial t} \left(\hat{U}_I(t) |\psi(0)\rangle \right) = \hat{V}_I(t) \hat{U}_I(t) |\psi(0)\rangle$$

$$\boxed{i \frac{d}{dt} \hat{U}_I(t) = \hat{V}_I(t) \hat{U}_I(t)} \quad U_I(t) = U_0^\dagger U_S$$

In a small step dt

$$U_I(dt) = \hat{I} - i \hat{V}(t) dt + \mathcal{O}(dt^2)$$

$$\hat{U}_I(t_N, t_0) = \hat{U}_I(t_N, t_{N-1}) \dots \hat{U}_I(t_1, t_0)$$

$$\bullet dt = t_{k+1} - t_k$$

Carry out same analysis we did for time evolution operator w/ $\hat{H} \rightarrow \hat{V}_I$

$$\begin{aligned} \hat{U}_I(t_N, t_0) &= \left(\mathbb{1} - i \hat{V}_I(t_{N-1}) dt + \dots \right) \dots \left(\mathbb{1} - i \hat{V}_I(t_0) dt + \dots \right) \\ &\xrightarrow[N \rightarrow \infty]{t=t_0} \mathbb{1} + (-i) \int_{t_0}^t dt_1 \hat{V}_I(t_1) + (-i)^2 \int_{t_0}^t dt_2 \int_{t_0}^{t_2} dt_1 \hat{V}_I(t_2) \hat{V}_I(t_1) + \\ &\quad + (-i)^m \int_{t_0}^t dt_m \int_{t_0}^{t_m} dt_{m-1} \dots \int_{t_0}^{t_2} dt_1 \hat{V}_I(t_m) \dots \hat{V}_I(t_1) + \dots \end{aligned}$$

$\mathcal{T} \rightarrow$ time ordered product as before:

$$\begin{aligned} \hat{U}_I(t) &= \sum_{n=0}^{\infty} \frac{(-i)^n}{n!} \int_{t_0}^t dt_1 \dots \int_{t_0}^{t_n} dt_n \mathcal{T} \{ \hat{V}_I(t_1) \dots \hat{V}_I(t_n) \} \\ &= \mathcal{T} \left\{ \exp \left[-i \int_{t_0}^t dt' \hat{V}_I(t') \right] \right\} \end{aligned}$$

Dyson series.

Show: $\hat{U}_I(t, t_0) = e^{i\hat{H}_0 t} U(t, t_0) e^{-i\hat{H}_0 t_0}$

using $|\psi(t)\rangle_S = \hat{U}(t, t_0) |\psi(t_0)\rangle$

and $|\psi(t)\rangle_I = \hat{U}_I(t, t_0) |\psi(t_0)\rangle_I$

Proof:

$$\begin{aligned} |\psi(t)\rangle_I &= \hat{U}_I(t; t_0) |\psi(t_0)\rangle_I \\ &= \hat{U}_I(t; t_0) e^{+i\hat{H}_0 t_0} |\psi(t_0)\rangle_S \end{aligned}$$

But also

$$\begin{aligned} |\psi(t)\rangle_I &= e^{+i\hat{H}_0 t} |\psi(t)\rangle_S \\ &= e^{+i\hat{H}_0 t} \hat{U}(t; t_0) |\psi(t_0)\rangle_S \\ \Rightarrow U_I(t; t_0) &= e^{+i\hat{H}_0 t} \hat{U}(t; t_0) e^{-i\hat{H}_0 t_0} \end{aligned}$$

Take $|i(t)\rangle = U_I(t) |\psi(0)\rangle = \sum |n\rangle \langle n| U_I(t; t_0) |i\rangle$

n.b. $\langle n | U_I(t; t_0) | i \rangle = e^{+i(E_n t - E_i t_0)} \langle n | U(t, t_0) | i \rangle$

$$\Rightarrow |\langle n | U_I(t; t_0) | i \rangle|^2 = |\langle n | U(t; t_0) | i \rangle|^2$$

$$|i(t)\rangle_I = U_I(t) |i\rangle$$

$$= \sum_n C_n(t) |n\rangle$$

$\hookrightarrow \langle n | U_I(t) | i \rangle$

Compare

$$|i(t)\rangle_S = \sum_n U(t) |n\rangle \langle n | i \rangle$$

$$= \sum_n e^{-iE_n t} C_n |n\rangle \quad \text{for } t\text{-indep. } H_0$$

$$|i(t)\rangle_I = e^{i\hat{H}_0 t} |i(t)\rangle_S$$

$$\Rightarrow |i(t)\rangle_S = \sum_n C_n(t) e^{-iE_n t} |n\rangle$$

$$C_n^{(0)}(t) = \delta_{ni} \quad \text{indep of } t$$

$$C_n^{(1)}(t) = -i \int_0^t \langle n | V_I(t') | i \rangle dt'$$

$$= -i \int_0^t e^{i\omega_{ni} t'} V_{ni}(t') dt'$$

$$\omega_{ni} = E_n - E_i$$

Same as before.

$$C_n^{(2)}(t) = (-i)^2 \sum_m \int_0^t dt_2 \int_0^{t_2} dt_1 e^{i\omega_{nk} t_2} V_{nk}(t_2) e^{i\omega_{ki} t_1} V_{ki}(t_1)$$

⋮

Propagators:

$$K_I(x_2, t_2; x_1, t_1) = \langle x_2 | \hat{U}_I(t_2, t_1) | x_1 \rangle$$

$$\Rightarrow \Psi_I(x_2, t_2) = \int dx_1 K_I(x_2, t_2; x_1, t_1) \Psi_I(x_1, t_1)$$

$$|\Psi_I(t)\rangle = U_I(t, t_0) |\Psi_I(t_0)\rangle$$

$$K_I(x_2, t_2; x_1, t_1) = \int dx K_0(x_2, t_2; x, t_1) K_I(x, t_1; x_1, t_1)$$

$$K_I(x_2, t_2; x_1, t_1) = \langle x_2 | U_I(t_2, t_1) | x_1 \rangle$$

Dyson series:

$$K_I = \langle x_2 | U_I | x_1 \rangle \equiv \langle x_2, t_2 | x_1, t_1 \rangle$$

$$= \delta(x_2 - x_1) + \sum_{n \geq 1} (-i)^n \int_{t_1}^{t_2} dt'_n \dots \int_{t_1}^{t_2} dt'_1 \langle x_2 | T(V_I(t'_n) \dots V_I(t'_1)) | x_1 \rangle$$

If we have a local potential:

$$\hat{V}(t) = \int dx |x\rangle V(x, t) \langle x|$$

$$\Rightarrow \langle x_2 | \hat{V}(t) | x_1 \rangle = V(x, t) \delta(x_2 - x_1)$$

Then, we can rewrite: